PRACTICAL WORKSHEET – EXPERIMENT

Experiment Title

PL/SQL Package Development: Procedures and Shared Cursors for Enterprise Applications

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1. Aim of the Session

The aim of this practical session is to design, implement, and execute a PL/SQL package in Oracle that includes procedures and shared cursors to achieve modular, reusable, and efficient database programming.

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2. Software Requirements

Database Management System:
Oracle / PostgreSQL

Database Administration Tool:
SQL Developer / pgAdmin

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3. Objective of the Session

After completing this practical, the learner will be able to:
· Understand the concept and structure of PL/SQL packages
· Differentiate between package specification and package body
· Develop packages containing procedures and shared cursors
· Implement modular and reusable database programs
· Use cursors for retrieving and processing multiple records
· Apply industry-relevant database development techniques

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4. Problem Statement

In enterprise applications, related database operations should be grouped together for better organization, performance, and reuse.
Design and implement a PL/SQL package that:
· Contains procedures to fetch and display employee details

· Uses a shared cursor to retrieve employee records
· Demonstrates execution of package procedures

(Industry relevance: Rippling, Oracle, SAP, PayPal)

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5. Practical / Experiment Steps

1. Create an Employee table with attributes such as emp_id, emp_name, and salary.
2. Insert sample employee records for testing.
3. Create a package specification (emp_package).
4. Declare a shared cursor to retrieve employee data.
5. Declare procedures to:
   · Display all employees
   · Fetch a specific employee
6. Create the package body implementing the logic.
7. Use cursor operations (OPEN, FETCH, CLOSE).
8. Use DBMS_OUTPUT.PUT_LINE to display output.
9. Handle exceptions using NO_DATA_FOUND.
10. Execute procedures using anonymous PL/SQL blocks.
11. Verify output and correctness.
12. Capture execution results for documentation.

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6. Procedure of the Practical

(i) Start the system and log in with valid credentials.
(ii) Open Oracle SQL*Plus / SQL Developer.
(iii) Enable output using SET SERVEROUTPUT ON.
(iv) Drop existing table and package if already present.
(v) Create the Employee table using SQL commands.
(vi) Insert sample records into the table.
(vii) Write the package specification with shared cursor and procedures.
(viii) Write the package body with implementation logic.
(ix) Compile the package successfully.
(x) Execute procedures using BEGIN...END blocks.

(xi) Display all employee records using cursor-based procedure.
(xii) Retrieve specific employee details using parameterized procedure.
(xiii) Observe output and verify correctness.
(xiv) Save code and capture screenshots.

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7. I/O Analysis (Input / Output Analysis)

Input

· SQL commands for creating Employee table
· Insert statements for sample data
· Package specification and package body
· Cursor-based procedure logic
· Anonymous PL/SQL blocks for execution
· Input values: emp_id

Output

· Display of all employee records using shared cursor
· Display of specific employee details (name and salary)
· Proper execution messages from PL/SQL
· Error message when employee record is not found

(Screenshots of execution and results are attached separately.)

```
     END;


ID: 1, Name: John, Salary: 50000
ID: 2, Name: Alice, Salary: 60000
ID: 3, Name: Bob, Salary: 55000


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.010
```

```
SQL> BEGIN
         emp_package.get_employee(2);
     END;


Name: Alice
Salary: 60000


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.006
```

```
SQL> CREATE OR REPLACE PACKAGE BODY emp_package AS

         ------------------------------------------------
         -- Procedure 1: Display all employees using cursor.
Show more...



Package Body EMP_PACKAGE compiled

Elapsed: 00:00:00.012
```

```
SQL> CREATE OR REPLACE PACKAGE emp_package AS

         -- Shared Cursor (visible to all procedures)
         CURSOR emp_cursor IS...
Show more...



Package EMP_PACKAGE compiled

Elapsed: 00:00:00.019
```

```
SQL> INSERT INTO employees VALUES (3, 'Bob', 55000)

1 row inserted.

Elapsed: 00:00:00.002
```

```
SQL> COMMIT

Commit complete.
```

```
1 row inserted.

Elapsed: 00:00:00.015
```

```
SQL> INSERT INTO employees VALUES (2, 'Alice', 60000)

1 row inserted.
```

```
SQL> CREATE TABLE employees (
         emp_id NUMBER PRIMARY KEY,
         emp_name VARCHAR2(50),
         salary NUMBER...
Show more...

Table EMPLOYEES created.

Elapsed: 00:00:00.014
```

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8. Learning Outcome

From this practical session, the student gained:

· Understanding of PL/SQL package structure (specification and body)
· Knowledge of shared cursors and their usage
· Ability to implement modular database programs
· Experience with cursor-based data retrieval
· Skills in exception handling using PL/SQL
· Understanding of enterprise-level database design
· Hands-on experience with Oracle PL/SQL programming
· Insight into real-world backend systems

9. Conclusion

The PL/SQL package was successfully created and executed using procedures and a shared cursor. The experiment demonstrated modular programming, efficient data retrieval, and reusable database logic, which are essential for enterprise-level applications.